ENVIRONMENT DIAGRAMS AND LAMBDA EXPRESSIONS Meta

COMPUTER SCIENCE MENTORS 61A

September 16 – September 19, 2025

Example Timeline

- Intros + Ice Breaker [5 min]

  Introduce yourself. Start off with an icebreaker of your choice!
- Mini-lecture: Environment Diagrams [7 min]

  Feel free to use the first question as an example/part of the mini-lecture.
- Q1: FooBar [7 min]
- Q2: ABCD [10 min]

  Feel free to spend more time on this if needed and skip the next question.
- Q3: Swap [If Time]
- Mini-lecture: Lambda Expressions and Higher Order Functions [7 min]
- Q4: Def to Lambda [5 min]
- Q5: Make Interval [7 min]
- Q6: Lambda Environment Diagram [If Time]

1 Environment Diagrams

1. Draw the environment diagram that results from running the code. What does the program print?

```
def foo(x):
    def bar():
        print(x)
    return bar

x = foo(3)
y = foo(4)
x()
y()
```

https://tinyurl.com/5n75xdk5

Created by Anaisha Das and Tavisha Patel. Worksheet feedback: forms.gle/q67BZrHHfoKbF6V28

Suggested Time: 7 min

- Go over this question as an example for the mini-lecture, if needed.
- Make sure to explan when new frames are opened and how parent frames work when calling x and y at the end.

2. Draw the environment diagram that results from running the following code.

```
a = 1
c = 2
def b(b):
    def d():
        return b + c
    return d()
c = b(a)
a = b(c)
```

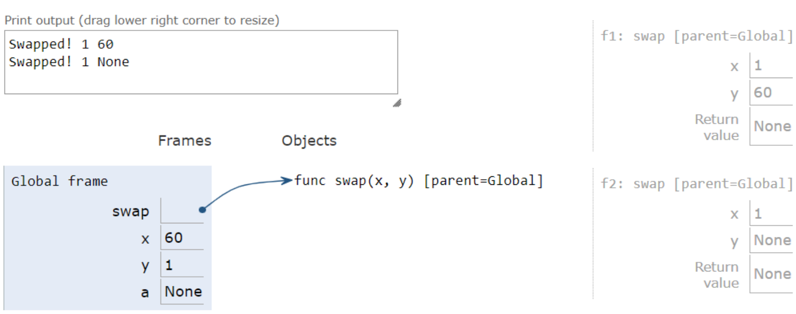
Suggested Time: 10 min

- The environment diagram questions can be a little long, so take extra time on this and skip Questions 3 and 6 if needed.

3. Give the environment diagram and console output that result from running the following code.

```
def swap(x, y):
    x, y = y, x
    return print("Swapped!", x, y)

x, y = 60, 1
a = swap(x, y)
swap(a, y)
```



https://tinyurl.com/y68m6qdj

Suggested Time: 10 min

- Optional, for if you have time after the first two questions or at the end.

2 Lambda Expressions and Higher Order Functions

1. Express the following expressions using lambdas instead of their named counterparts.

   (a) `square(4)`

   ```
   (lambda x: x * x)(4)
   ```

   (b) `sum_of_squares(3, 4)`

   ```
   (lambda x, y: x * x + y * y)(3, 4)
   ```

   (c)
   ```
   def hello_world():
       return 'Hello, world!'
   hello_world()
   ```

   ```
   (lambda: 'Hello, world!')()
   ```

   Suggested Time: 5 min

   - This is a quick introduction to lambda functions.
   - Make sure to highlight the structure of lambda expressions and how they are called.

2. Make a lambda function, `make_interval()`, that takes in the upper and lower bound of an interval, and returns a function that takes in a value `x` and checks whether `x` is in the interval [`lower`, `upper`], inclusive.

   ```
   >>> make_interval = ______________________________________
   >>> in_interval = make_interval(-1, 2)
   >>> in_interval(0)
   True
   >>> in_interval(61)
   False
   ```

   ```
   >>> make_interval = lambda lower, upper: lambda x: x <= upper and x >=
       lower
   >>> in_interval = make_interval(-1, 2)
   >>> in_interval(0)
   True
   >>> in_interval(61)
   False
   ```

   Teaching Tips

   - Question students exactly the function's purpose is- if they are stuck, walk them through the doctest and ask leading questions based on each one.
   - It is often easier for students to understand currying if you show the equivalent standard Python higher-order function (with def and nested functions).
   - It can help to explain the first functional arguments as "setting the bounds", and then subsequent arguments as checking those bounds.

Suggested Time: 7 min

3. Draw the environment diagram that results from running the code.

```
def a(y):
    d = 1
    b = lambda x: y(x)
    e = lambda x: x(3)
    return e(b)
d = 5
a(lambda x: 4 - x + d)
```

https://goo.gl/9vxEwv

Suggested Time: 7 min

- Optional, for if you have time.
- This question is about how lambda expressions work in environment diagrams, so briefly go over how this works if you don't have time to complete the whole question.